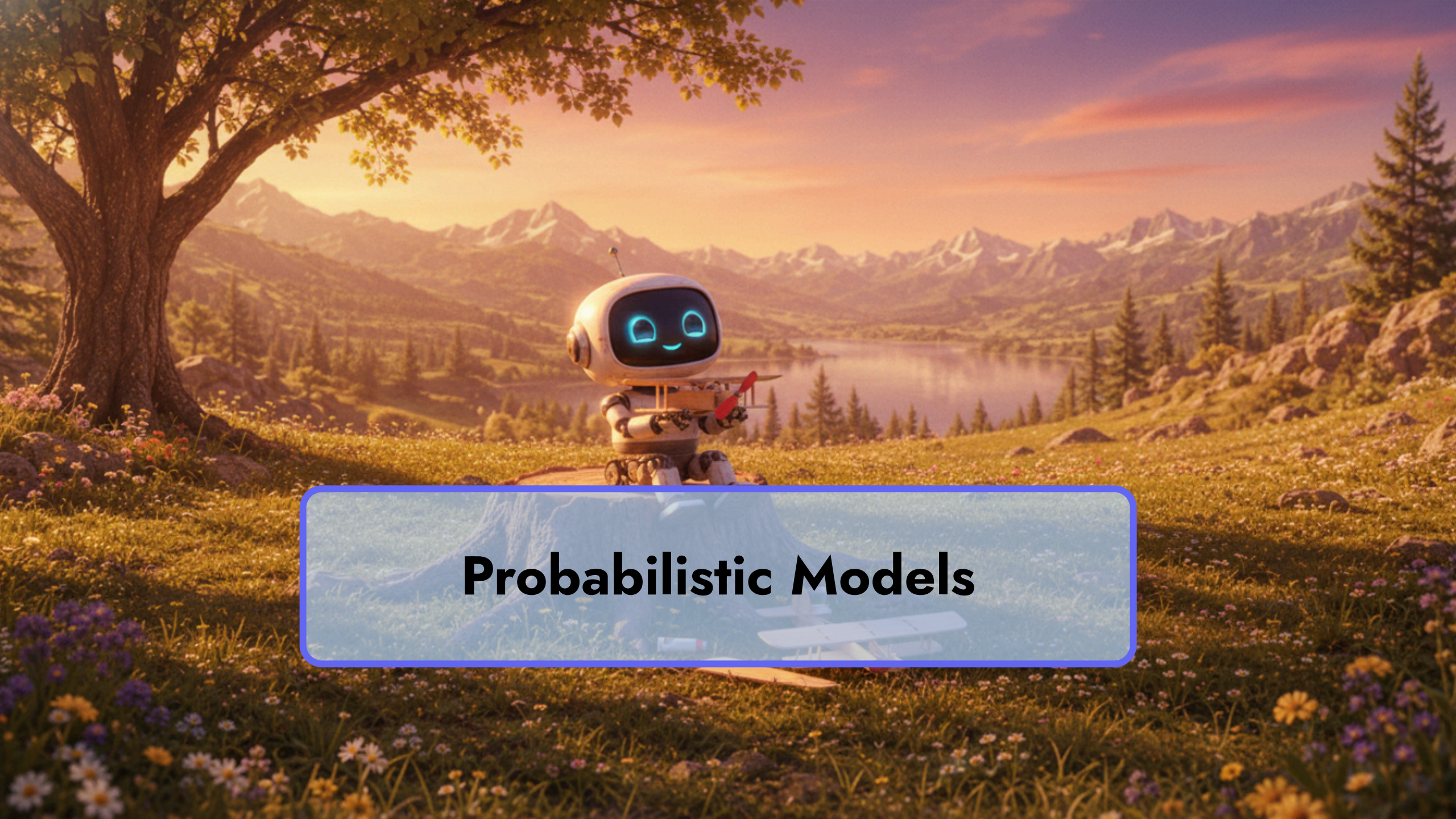

Probabilistic Models

Word Wall Demo

Interactive demo

Learning goals:

1. Feel comfortable with a probabilistic model

2. Understand how models can generate.

3. Run the story backwards to answer questions





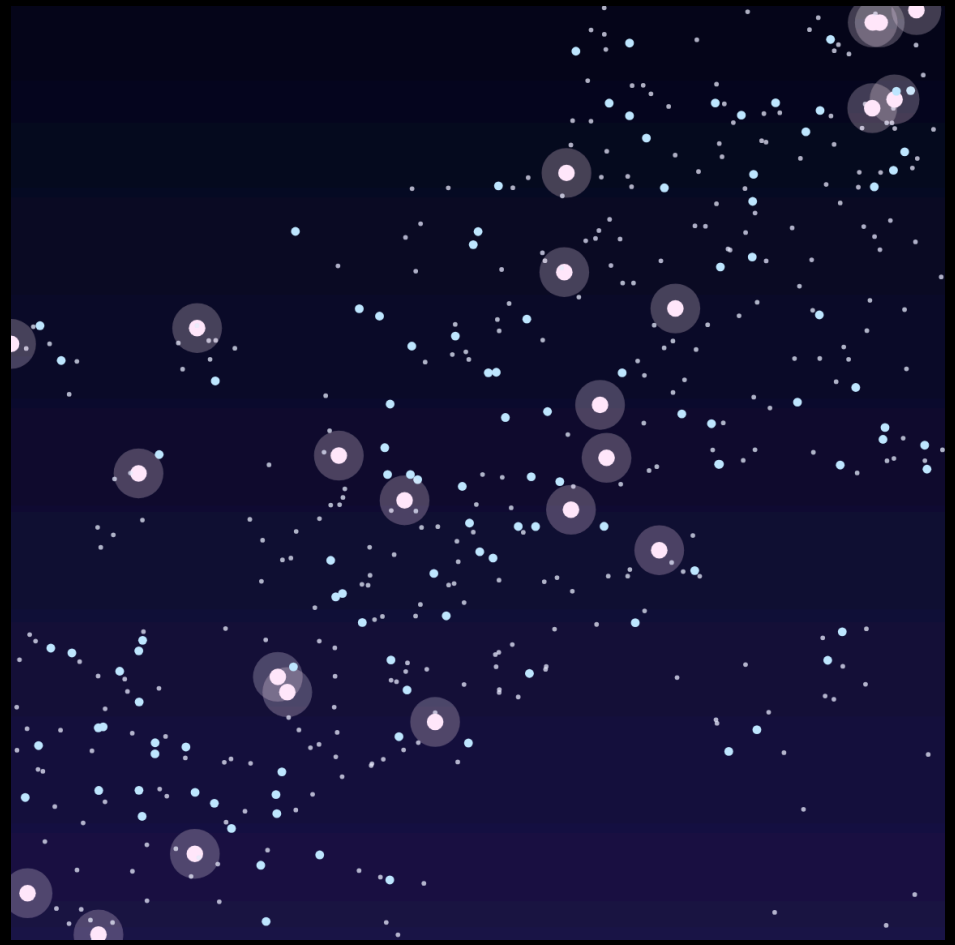




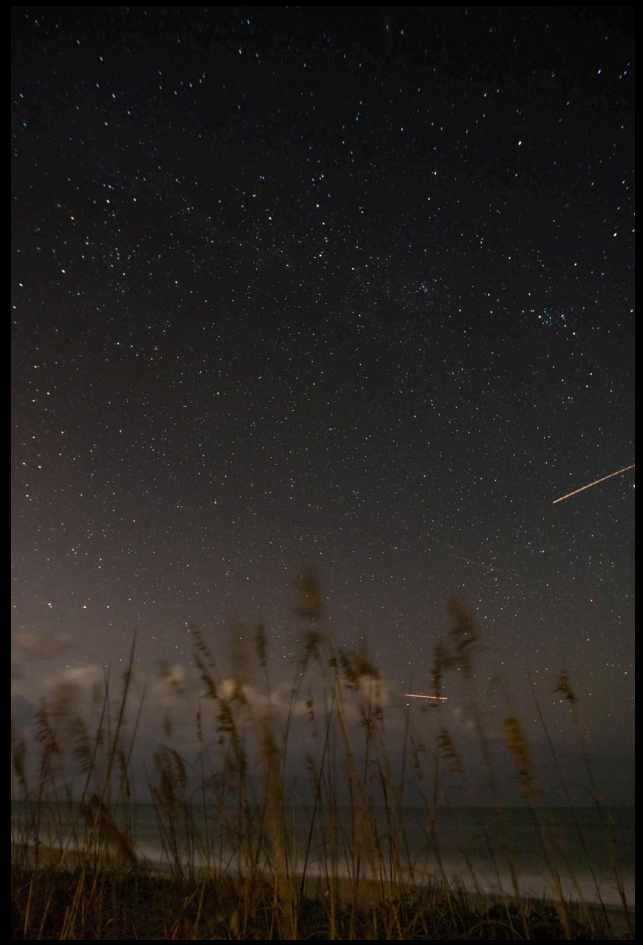
"Master small models, so that you can understand Large ones"

-- Chris Piech. Today. In this Lecture





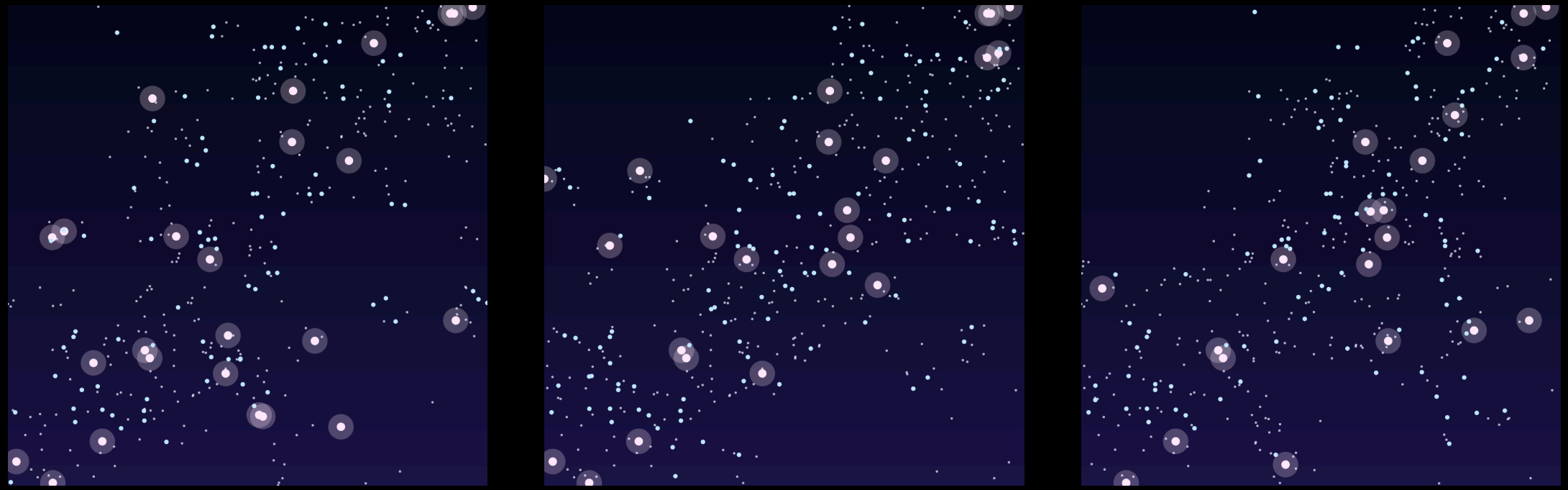
the story



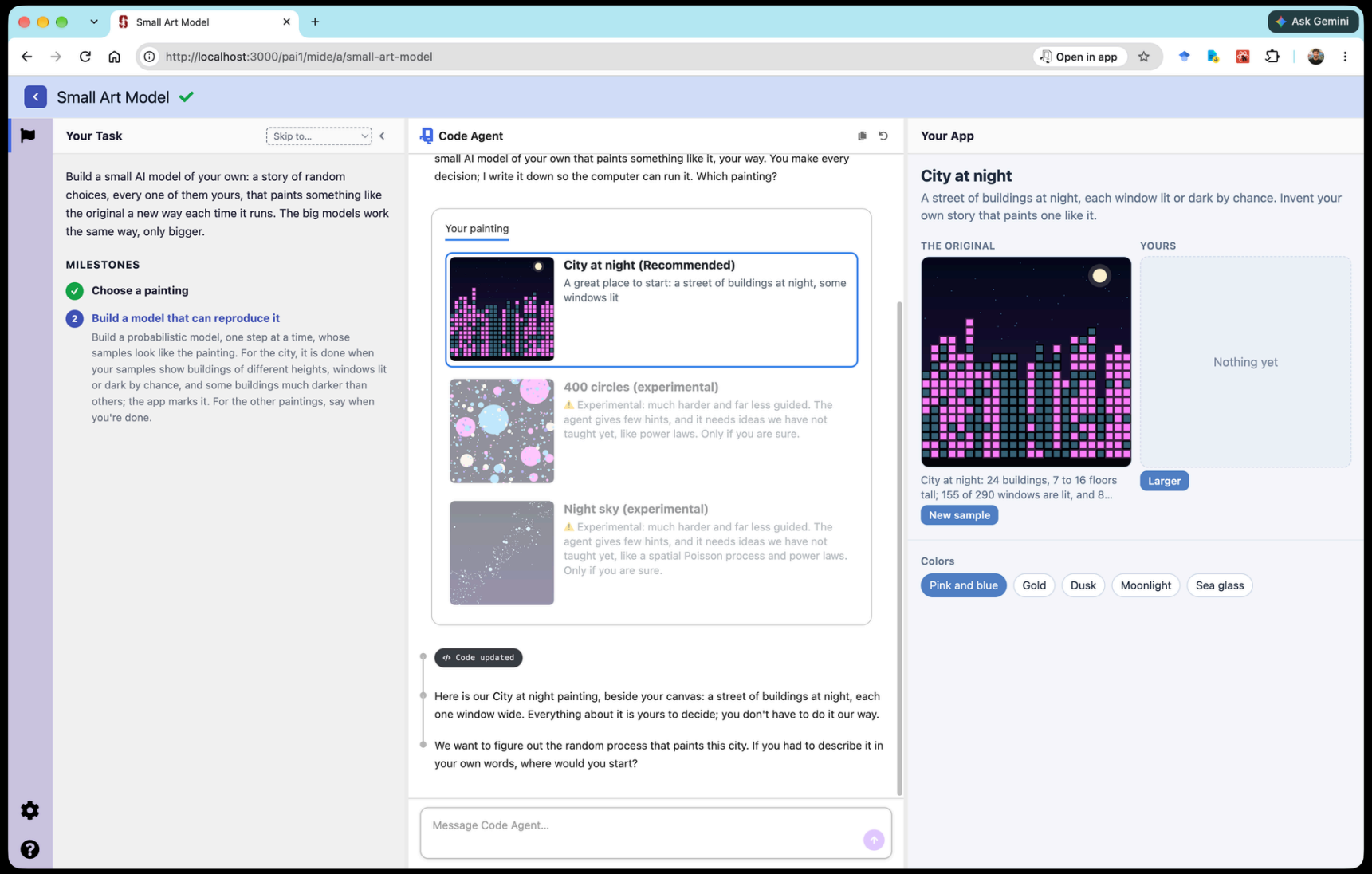
the real sky

Turn the knobs until the story's sky looks like the real one: learning by eye. A computer can do that from data. How it knows which way to turn each knob is next week. Photo: NASA/KSC, public domain.

None of these skies exist



Three parameters made all of them. What could the knobs be?

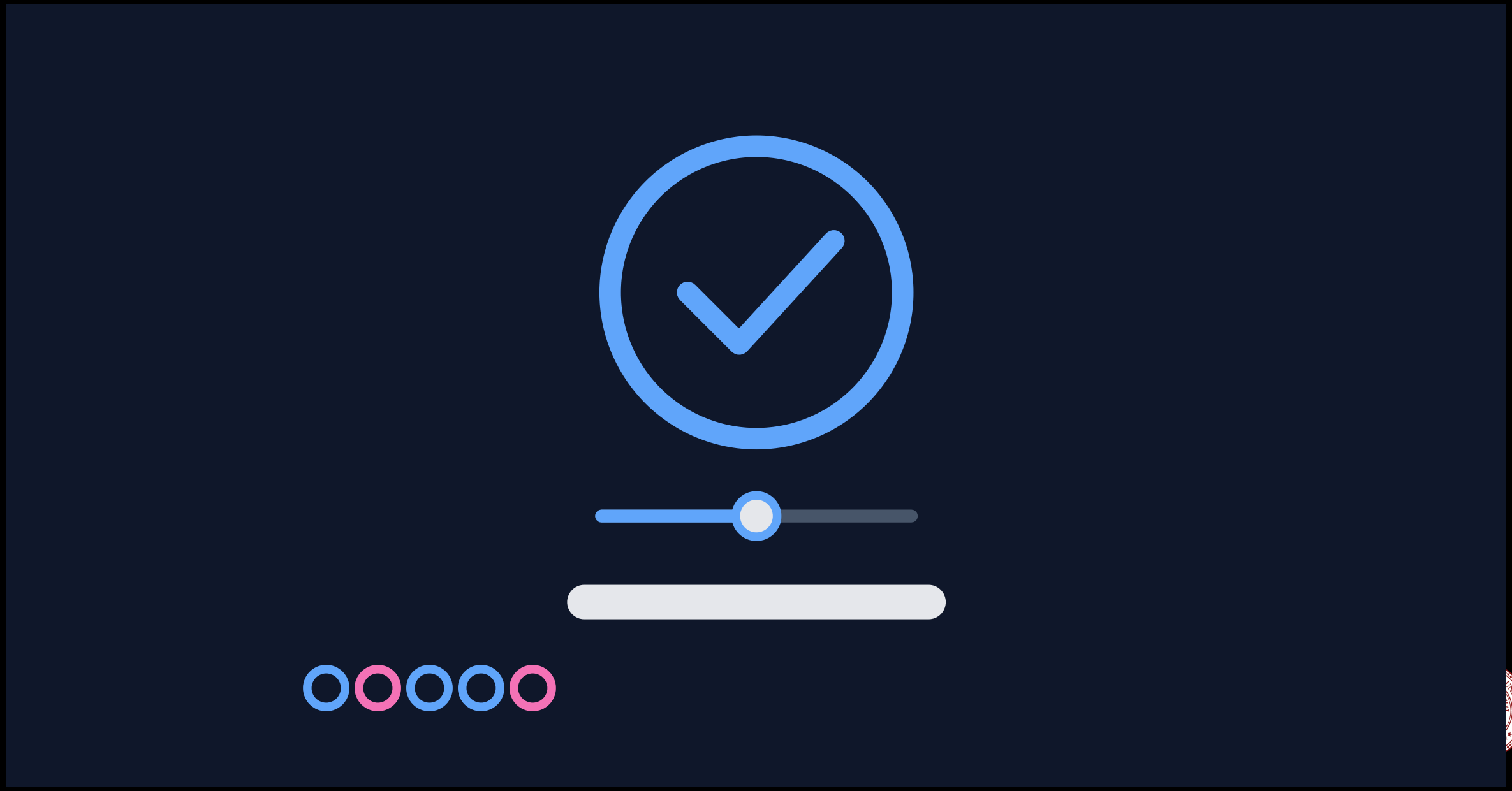


After this lesson you will be ready to do your assignment

Review


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1891

Bernoulli



Binomial

Flip a coin n times.

Probability p of success on each flip.



4 heads

Poisson



End Review


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DIE LUFT DER FREIHEIT WEHT
1891



A model is a probabilistic story with knobs

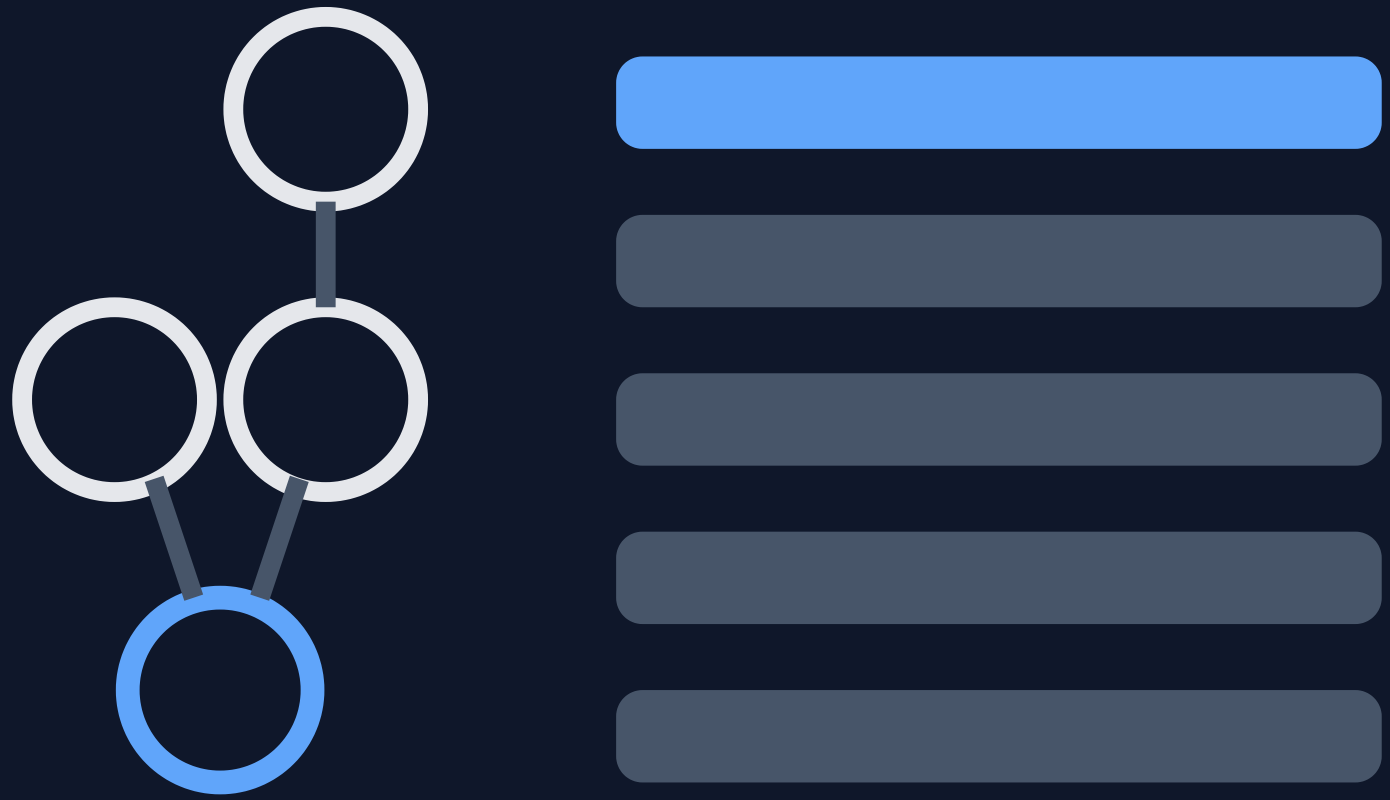




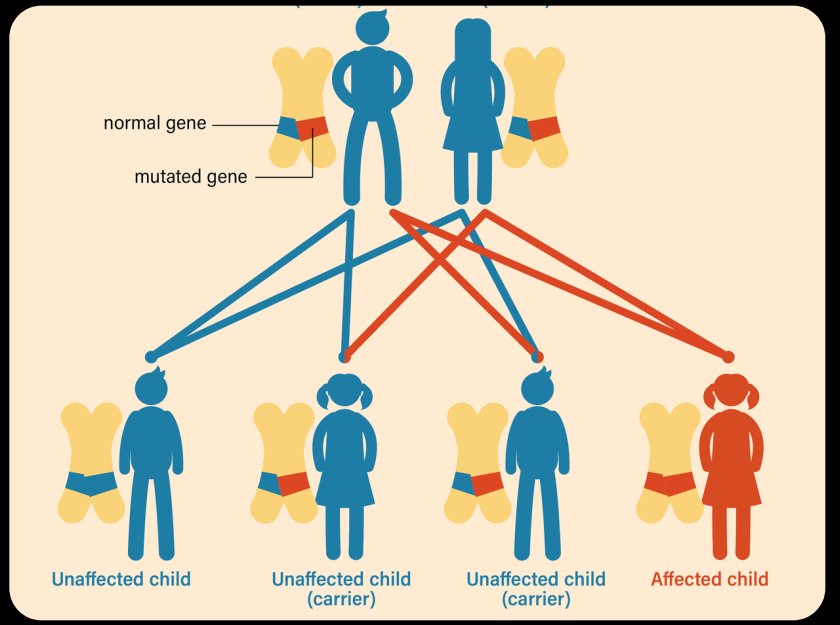
A model is a probabilistic story with ~~knobs~~

Parameters

A story that makes bats

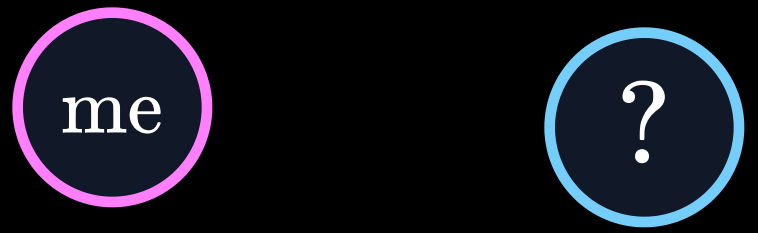





normal gene
mutated gene
Unaffected child
Unaffected child (carrier)
Unaffected child (carrier)
Affected child

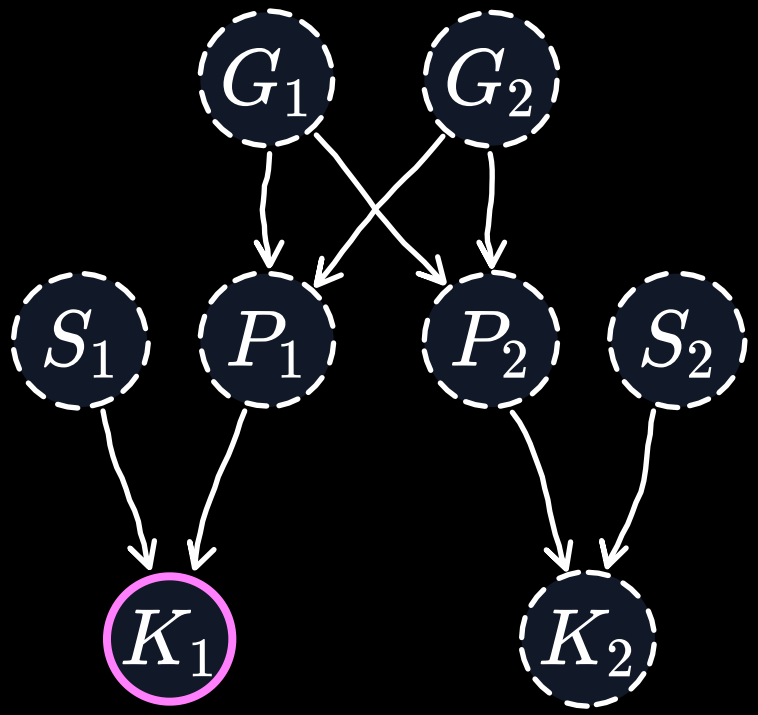
I carry a scary gene.

What is the probability that my cousin carries it too?



Out in the world, 1 person in 20 carries it. Is my cousin 1 in 20?

The family, one person at a time



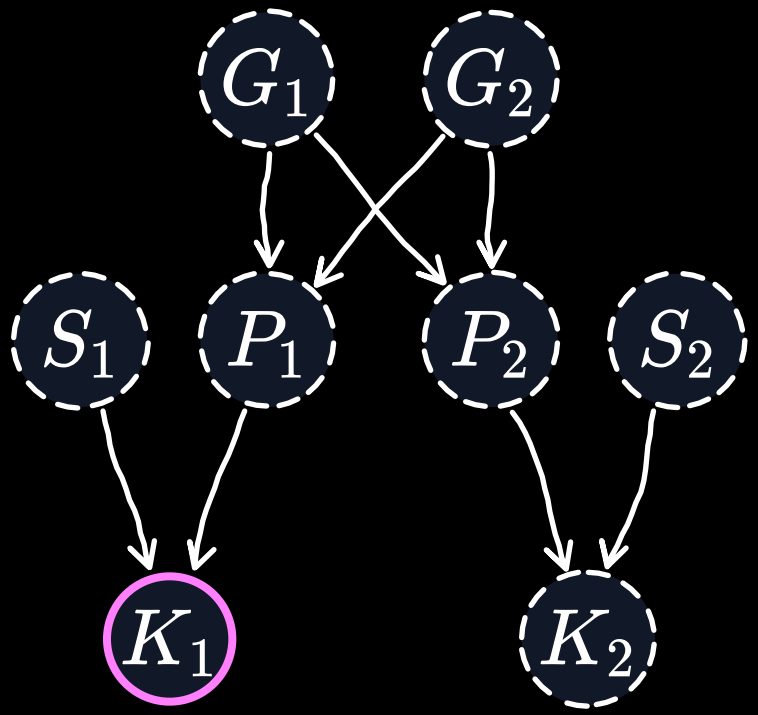
Gene rate: θ_1 0.05

Passed rate: θ_2 0.5

One family

Grandparent 1	**yes**
Grandparent 2	no
Parent 1	**yes**
Parent 2	**yes**
Spouse 1	no
Spouse 2	no
Cousin 1	**yes**
Cousin 2	no

Run the story 10,000 times



Gene rate: θ_1 0.05

Passed rate: θ_2 0.5

cousins.csv

G1	G2	P1	P2	S1	S2	K1	K2
True	False	True	True	False	False	True	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
True	False	False	True	False	False	False	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
True	False	True	False	False	False	True	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
False	True	True	True	False	False	True	True
False	False	False	False	True	False	False	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
False	True	True	True	False	False	True	True
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False

10,000 lines in all



Count the file

K_1 : I carry the gene
K_2 : my cousin carries it

$$\mathrm{P}(K_2 \mid K_1) \approx \frac{\mathrm{count}(K_2 \text{ and } K_1)}{\mathrm{count}(K_1)}$$
$$\approx 0.17$$

cousins.csv

G1	G2	P1	P2	S1	S2	K1	K2
True	False	True	True	False	False	True	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
True	False	False	True	False	False	False	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
True	False	True	False	False	False	True	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
False	True	True	True	False	False	True	True
False	False	False	False	True	False	False	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
False	True	True	True	False	False	True	True
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False
False	False	False	False	False	False	False	False

10,000 lines in all





Sample from a model and
then ask questions of the data!

Rejection Sampling

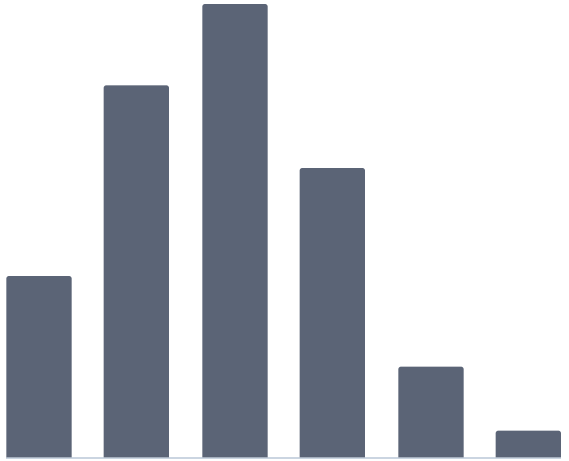
Ready to get creative?

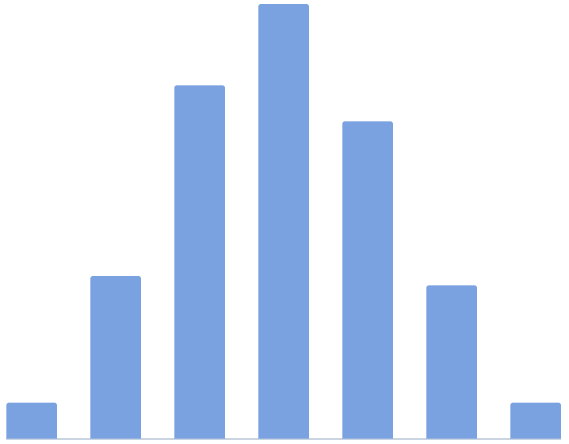


Code Agent

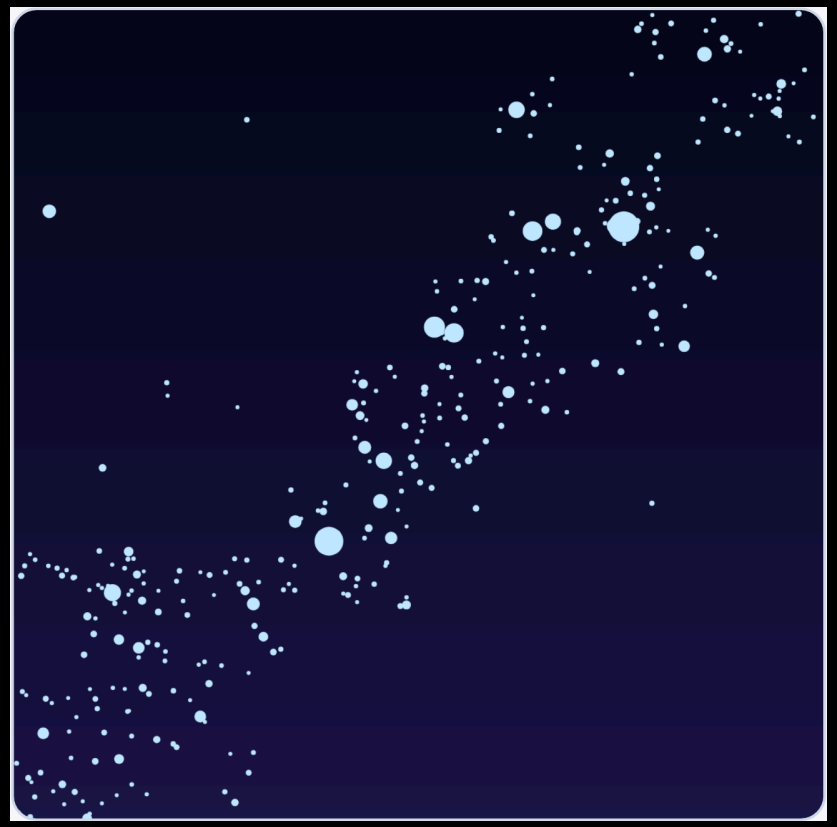
Model

Samples

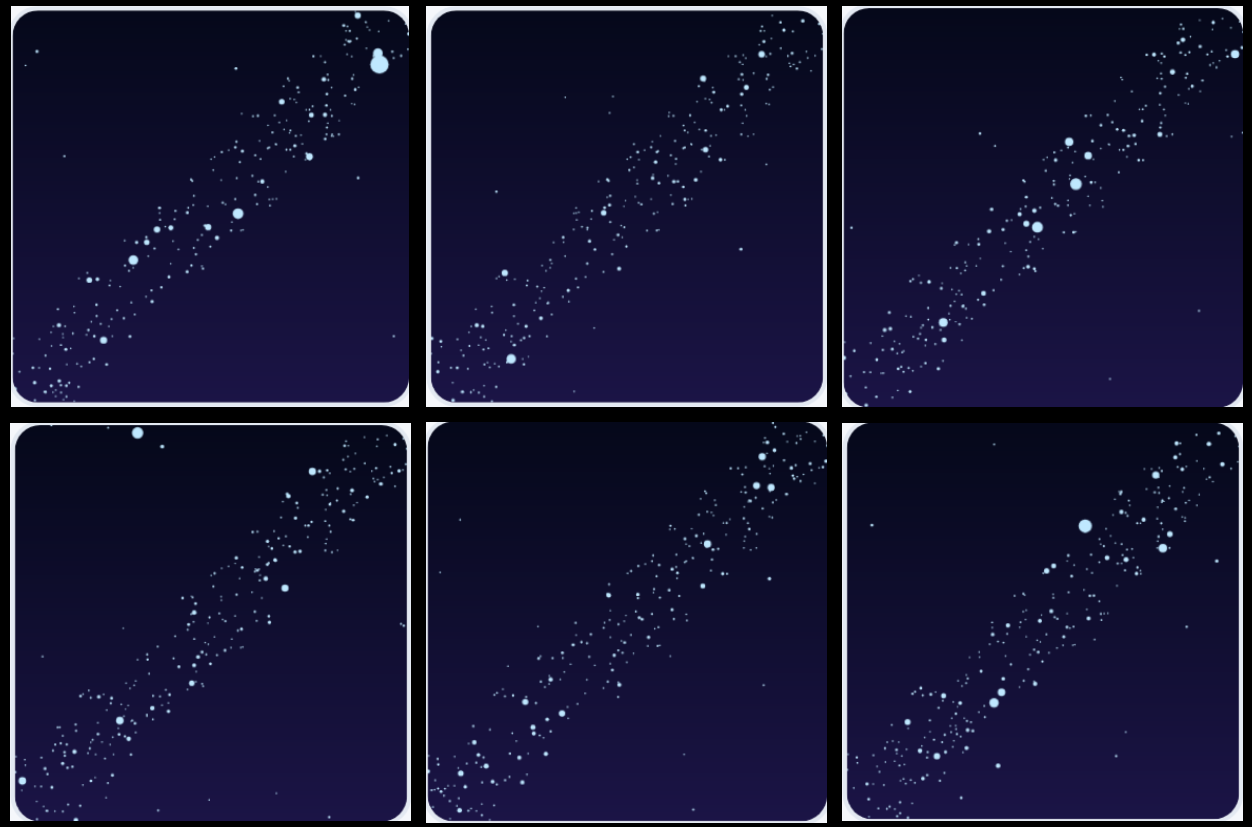




Seven skies from our model



A sky from our model



Six skies from our model



All models are wrong, some models are useful.

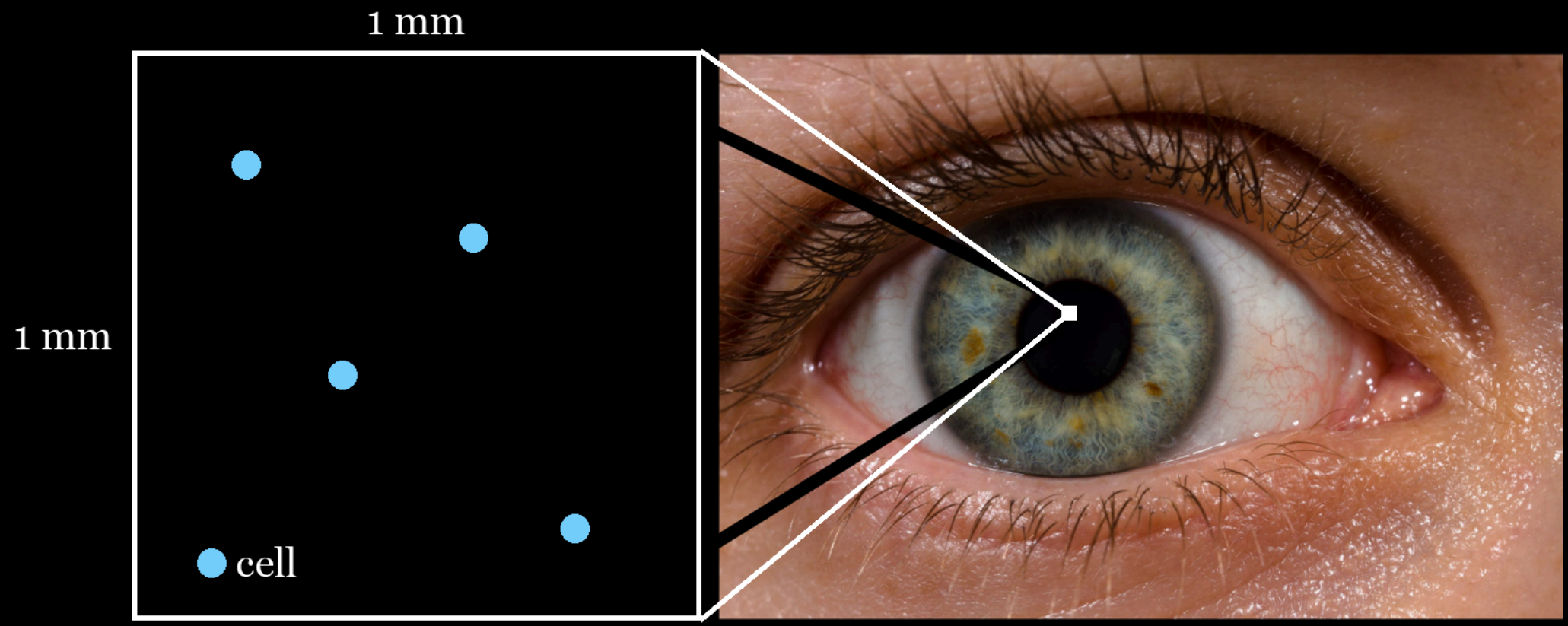
George Box. Ours draws a sky with a hard edge; it still says how many stars a patch should have.

The same skill builds a model of an eye disease.

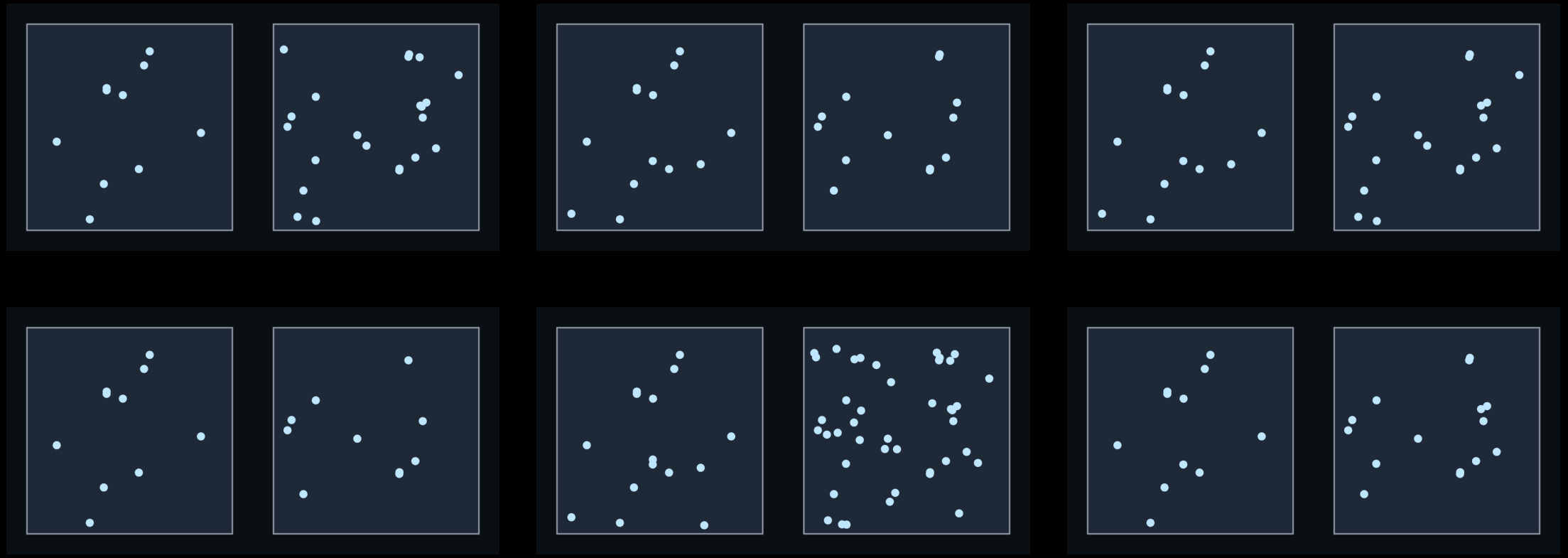


How the eye exam works

The doctor counts the cells in one tiny square.



Six eyes: a month ago, and today



Left: a month ago. Right: today. Each dot is a cell.

An open problem.

Join me in solving it during PAI.

is there a paper talking about how eye inflamation is a spation poisson (like grade 1, trace etc)

- I couldn't find one. I searched the web, PubMed and the full text of papers in Europe PMC, and no paper treats SUN anterior chamber cell grading (0 / 0.5+ "trace" / 1+ / 2+ …) as Poisson counts in the slit-beam volume, or works out how often that randomness puts an eye in the wrong grade. Where "Poisson" appears next to "anterior chamber cells", it's always a regression model. For example, Manni et al. (Ophthalmol Sci 2026) fit a Poisson mixed-effects model to cells counted on OCT scans. Nobody uses it to describe where the cells sit in the eye.

do you agree it should be a spatial poisson?

- Yes, as long as the claim is: given how inflamed the eye is, the number of cells in one look is Poisson. The argument for it is strong. What actually varies is the rate, not the Poisson form.

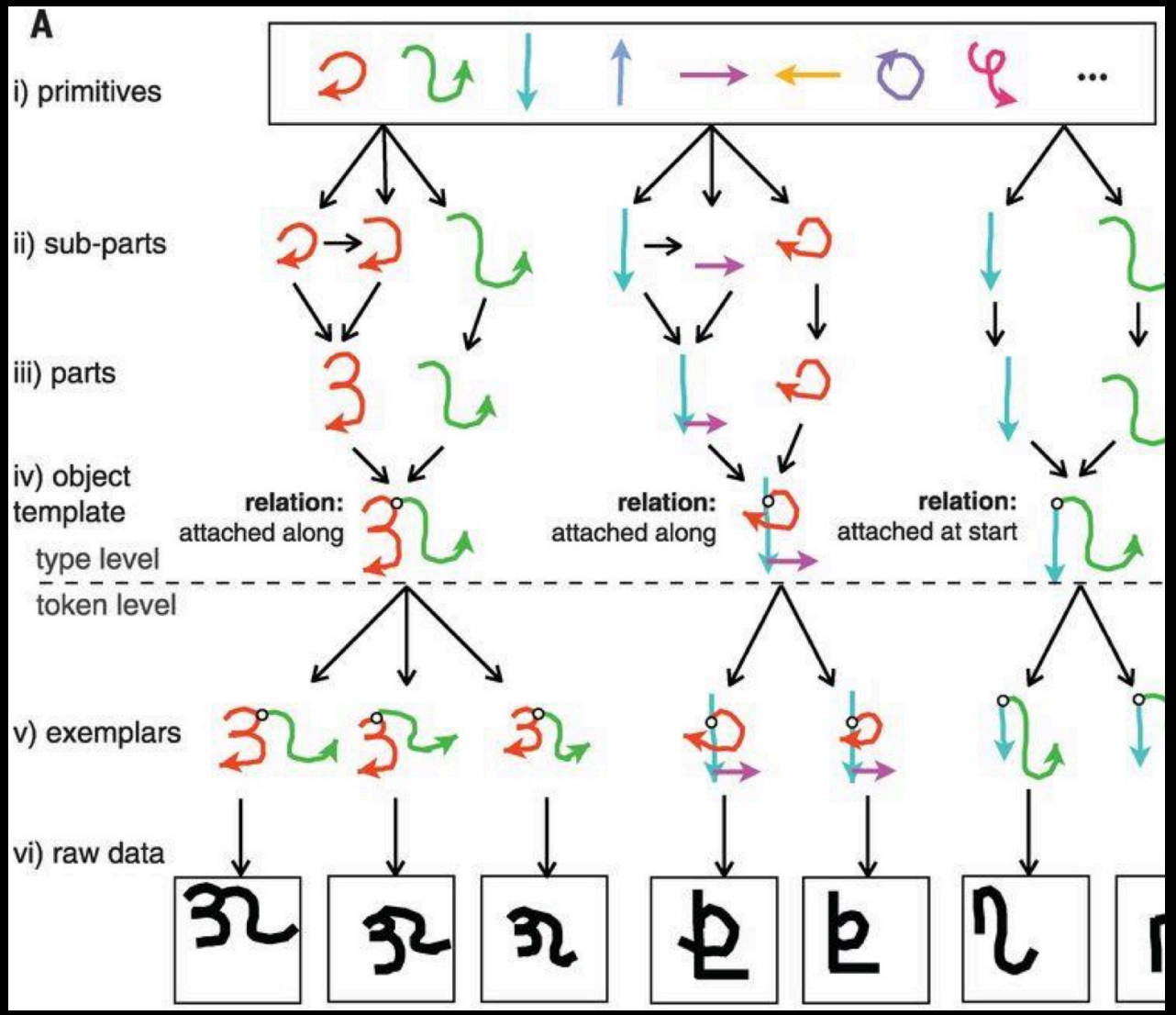
  Why Poisson is the right default

  - **Rare events.** The anterior chamber holds roughly 200 mm³ of fluid. A 1×1 mm beam through about 3 mm of chamber depth lights roughly 3 mm³, or 1–2% of it. If N cells drift around independently, the count in the beam is Binomial(N, p) with N large and p small, which is Poisson(λV). This is the same argument Student made for yeast in hemocytometer squares.
  - **Cells don't crowd each other.** Even at grade 4+ (more than 50 cells in about 3 mm³), cells about 10 µm across fill a negligible share of the fluid. Nothing stops one cell from sitting near another, so counts in separate regions are independent. That's what makes it a *spatial* Poisson process.
  - **Missed cells don't break it.** If each cell is seen with probability q, the seen count is still Poisson, just with rate λqV. Dim light or a tired examiner lowers the rate but keeps the distribution.
  - **Each look is a fresh draw.** Heat currents keep stirring the fluid, so looks a few seconds apart are close to independent. That fits a "look again" beat in the lecture.

Ready to get creative?

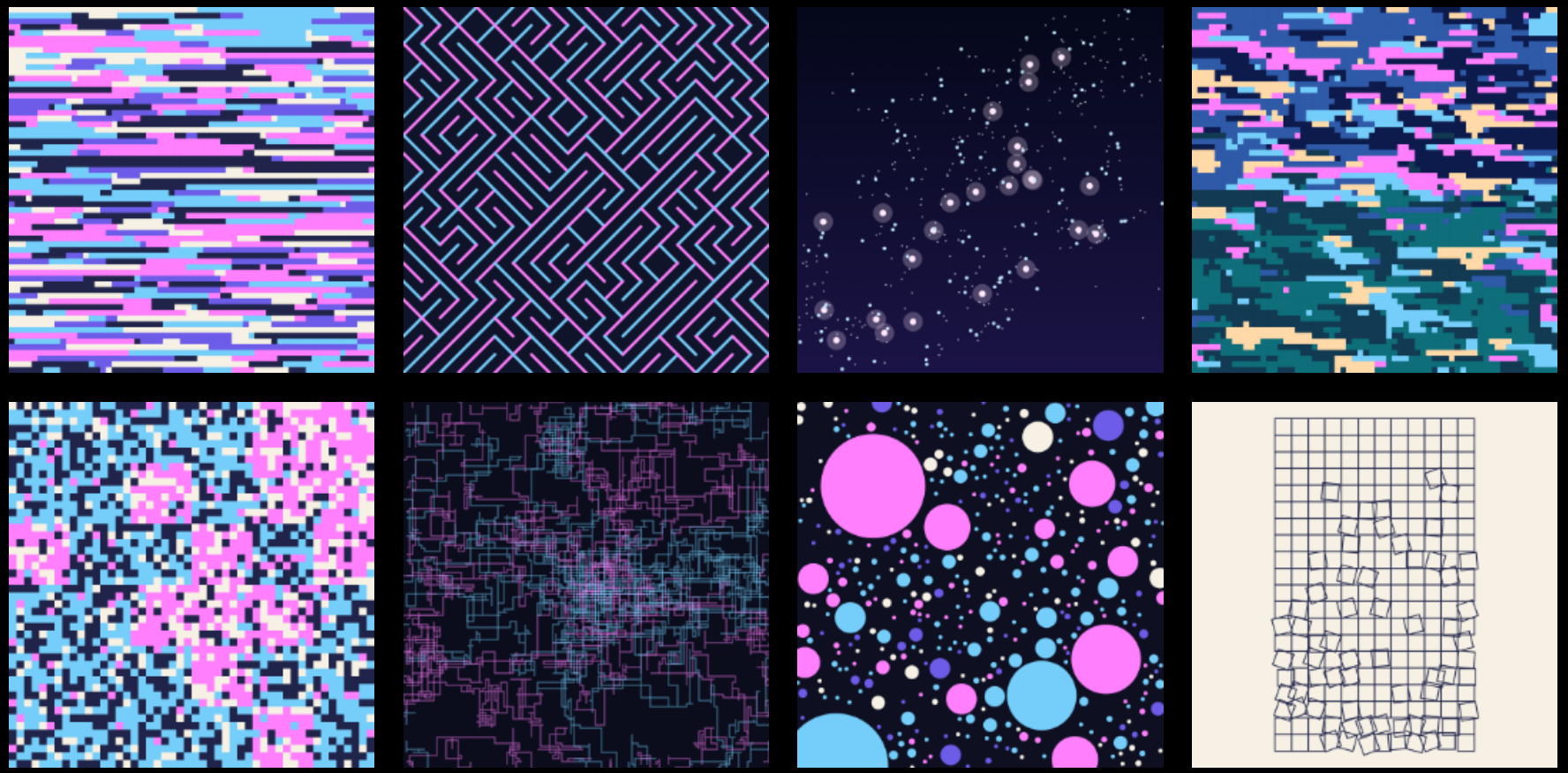


A story that writes letters



B. Lake, R. Salakhutdinov, J. Tenenbaum, "Human-level concept learning through probabilistic program induction," Science, 2015.

Your turn: a story that paints



Every one of these paintings is a story with knobs. This week you invent your own: Small Art Model.